

Comparative numerical and experimental analysis of thermal and hydraulic performance of improved plate fin heat sinks

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HIGHLIGHTS

- Performance analysis of the application of longitudinal perforations and slots in plate-fin heat sinks.
- Simultaneous improvement of thermal and hydraulic performance of plate-fin heat sinks.
- Reduction in weight of plate-fin heat sinks for heat transfer applications.
- Validation of numerical model with literature and experiments.
- Application of additive manufacturing tools for the fabrication of experimental components.

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ABSTRACT

The benefits of multiple perforations and slots in a plate-fin heat sink are investigated using a conjugate heat transfer model and complementary experimentation. Heat transfer and pressure drop are analyzed across the two, novel plate-fin heat sinks (with perforations and slots). The experimental data validates the conjugate heat transfer CFD model for the air-cooled heat sink. The results from the CFD model show that novel plate-fin heat sinks tend to have a higher heat transfer coefficient than the plane fins without slots and perforations. Also, pressure drop of the novel plate-fin heat sinks is lower than the plane plate-fin heat sink, so less fan power is required for novel heat sinks. For a range of Re number 13,049 to 52195, on average novel plate-fin heat sink-1 (NPFHS-1) and novel plate-fin heat sink-2 (NPFHS-2) have 42.8% and 35.9% higher Nu number than that of the plane plate-fin heat sink (PPFHS). An additional advantage of the reduction in mass of heat sink is achieved because of slots and perforations.

1. Introduction

Caloric theory proposed by Antonie Lavoisier demonstrated that heat is a massless fluid called caloric. This fluid when possessed by a body, increases its temperature and, when removed from a body, decreases its temperature [1]. Although caloric theory holds importance in physical understanding of heat, it was held down by James P. Joule's experiment, in 1843, which proved that heat is not a physical substance but a form of energy that is transferred from one body to another because of temperature gradient [2].

In most engineering systems, dissipation of the excess heat is important. The efficiency of the system may drop because of excess heat. An internal combustion engine converts only 25% of the power

generated, into useful mechanical work. Almost 30% of the generated power is wasted as excess heat, which needs to be dissipated through cylinder walls [3]. There are multiple coolants used for heat transfer. For example, in automotive radiators water, ethylene glycol or oils are used. While nanofluids are thermally more effective, they aren't sustainable. So, hybrid nanofluids (thermally efficient nanoparticles and sustainable nano particles) can be used in automobile radiators [4]. The components of an integrated circuit (e.g., transistors, resistors, inductors, capacitors) produce heat which affects the performance and lifespan of electronic equipment. A uniform temperature distribution across the air-cooled photovoltaic panels is vital. Also, the photovoltaic modules perform better under lower temperatures [5].

To overcome overheating, fins are used. Fins are extended surfaces

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Nomenclature		Symbols	
CFD	Computational Fluid Dynamics	H	Height of fins
PFHS	Plate-Fin Heat Sink	L	Length of fins
PPFHS	Plain Plate-Fin Heat Sink	W	Width of heat sink
NPFHS-1	Novel Plate-Fin Heat Sink-1	s	Distance between consecutive fins
NPFHS-2	Novel Plate-Fin Heat Sink-2	t	Thickness of fins
PCM	Phase Change Material	t _{base}	Thickness of the heat sink base
NePCMs	Nanoparticles enhanced Phase Change Materials	t _p	Thickness of the perforation
BCC	Body Centered Cubic	t _s	Thickness of novel fins at slotted location
DMLS	Direct Metal Laser Sintering	H _c	Height of the perforation channel
ALM	Additive Layer Manufacturing	$Re = \frac{\rho_{air} U_{air} D_h}{\mu_{air}}$	Reynold's number
SLM	Selective Laser Melting	ρ_{air}	Density of air
RANS	Reynolds-Averaged Navier-Stokes	U	Speed of air
SST	Shear Stress Transport	$D_h = \frac{4A_C}{P}$	Hydraulic diameter of the test section
SIMPLE	Semi Implicit Method for Pressure Linked Equations	μ	Dynamic viscosity of air
DC	Direct Current	A _C	Cross-sectional area of the test section
AC	Alternating Current	P	Perimeter of the test section
FDM	Fused Deposition Modeling	k _s	Conduction coefficient
PFHS	Plate-Fin Heat Sink	T	Temperature
PPFHS	Plain Plate-Fin Heat Sink	C _p	Specific heat at constant pressure
NPFHS-1	Novel Plate-Fin Heat Sink-1	Pr	Prandtl number
NPFHS-2	Novel Plate-Fin Heat Sink-2	Q	Rate of heat transfer
PCM	Phase Change Material	Pk	production limiter of turbulence
NePCMs	Nanoparticles enhanced Phase Change Materials	S	Invariant strain rate
BCC	Bodies Centered Cubic	F	Blending function
DMLS	Direct Metal Laser Sintering	Δx	Insulation thickness
ALM	Additive Layer Manufacturing	h _p	Convection coefficient calculated using projected area of heat sink
SLM	Selective Laser Melting	$Nu_p = \frac{h_p L}{k_{air}}$	Nusselt number calculated using projected area of heat sink
RANS	Reynolds-Averaged Navier-Stokes	ΔP	Pressure-drop
SST	Shear Stress Transport	f	Friction factor
SIMPLE	Semi Implicit Method for Pressure Linked Equations	m	mass
DC	Direct Current	V	Volume
AC	Alternative Current		
FDM	Fused Deposition Modeling		

or bluff plates, having a large surface area for enhancement of convection heat transfer [6,7]. Many modifications have been made in the geometry of fins, like dimples, cavities, slots, holes, grooves and microchannels. Such modifications in geometry provide more surface area for convection heat transfer, while reducing their material. Nanoparticles also augment the heat transfer rate in various applications. When TiO₂ nanofluid is used as working fluid in flat-plate solar collector, heat absorption rate increases with the increase of the volume fraction of the nanoparticles [8]. Similarly, a higher volumetric concentration of SiO₂/water-based nanofluids enhances the heat transfer rate [9].

Meinders et al. experimentally observed the enhancement in heat transfer rate by making cubical protrusions, for Re number 795–6656. Their research showed that, when position of the cubical protrusions is changed relative to each other, the rate of heat transfer changes. He also used liquid crystal thermography and oil film visualization to determine the flow behavior of fluid over the heat sink [10]. Nakamura et al. studied the variation of heat transfer rate through a surface having cubical protrusions for a range of Re number 4.2×10^3 – 3.3×10^4 [11]. Ničeno et al. numerically studied the variation in the heat transfer rate through a surface containing cubical protrusions. In this case, each cube was composed of multiple layers of matrix. The flow across the protrusions was considered turbulent and behavior of fluid flow and vortex generation in the domain was observed [12]. Sara studied the heat transfer rate through a flat plate and the effect of a attaching a solid rectangular block and perforated rectangular block to the flat plate, on its thermal performance. The results showed that rectangular block enhanced the heat transfer rate through flat plate, and perforated

block exhibited the best heat dissipation rate [13].

Yu et al. numerically and experimentally analyzed the thermal resistance of the plate-fins heat sink and the heat sink containing plate and pin fins. Comparison indicated that plate fins have 30% less thermal resistance than the heatsink containing both plate and pin fins [14]. Li et al. studied the effect of various widths and lengths of plate fins on their heat transfer rate, at different Re numbers, under confined impinging jet conditions, using infrared thermography. The reduction in thermal resistance of plate fins was more obvious for increased width than increased length of the fins [15]. Wei et al. studied the thermal performance of dimpled fins and pressure drop across them, on the flat tube. A comparison between plane fins and dimpled fins showed that dimpled fins have better heat transfer performance [16]. Chang enhanced the thermal performance of a vertical array of dimpled fins under natural convection. Their numerical study also showed significantly higher Nu number of dimpled fins as compared to the smooth fins [17]. Phase change materials (PCMs) also prove to have better cooling effect. Furthermore, their combination with fins causes further enhancement in heat transfer rate. Usually PCMs have a low thermal conductivity. Combination of PCMs and nanoparticles, known as Nanoparticles enhanced Phase Change Materials (NePCMs), can also cause high cooling rate [18,19].

When fins are used in heat exchangers, heat transfer rate increases, but this introduces pressure loss in the system, which requires high pumping power. Also, fins increase the weight of the system by adding extra material which increases cost. Modern heat transfer applications require the fins to have compact and practical design. Especially in electronics, it is required for the heat sink to be lightweight while being

efficient. This can be achieved by using perforated and interrupted fins. Perforations enhance the heat transfer rate because they offer extra surface area for convection, while effectively reducing pressure loss and weight of the heat sink [20,21]. Bilen studied the effect of porosity by airflow through an aluminum porous-fin heat sink at a Re number 5000–35000, in a rectangular duct. It was established that higher Nu number and lower pressure drop are achieved at bigger gap heights [22]. Minichannels incorporated with fins can enhance the heat transfer rate. The nanofluids are also used in minichannels. While the dimensions of the channel affect the thermal performance, concentration of nanofluids (e.g. Al_2O_3) also affect the heat transfer and pressure-drop [23].

According to Jassem, when the diameter of the perforation is increased, the overall heat transfer rate increases. Different perforation shapes were experimentally analyzed, which showed that the shape of perforation is a significant factor that affects the heat transfer coefficient [24]. Gurav et al. investigated the effect of elliptical perforation on a rectangular plate fin through finite element analysis. It was found that the elliptical perforations enhanced the heat dissipation rate and reduced the material of rectangular fin [25]. The same experiment was performed using circular perforations, and in comparison to the solid rectangular fins, perforated plate fins exhibited a higher heat transfer coefficient and lower drop in pressure across the heat sink [26]. Studies has shown that if we increase the number of perforations, the heat transfer rate through the heat sink increases which results into lower temperature at fin tips [20,21]. Sahin et al. studied the temperature distribution and friction characteristics of square pin-fin with perforations. They reported that perforations, porosity, and slots improve the heat transfer rate of fins [27].

Molki et al. compared the performance of the perforated baffle plate with that of a solid baffle plate and found that perforations caused low flow resistance and high heat transfer [28]. Sparrow et al. analyzed the heat transfer rate of a surface containing multiple holes in a rectangular pattern, facing the airflow, at different pitch-to-diameter ratios and different Re numbers, which showed that bigger diameter perforations enhance heat transfer rate while reducing pressure loss of the fluid [29]. Dorignac experimentally determined the values of convection coefficient on multi-perforated plates and derived a correlation of heat dissipation for a perforated flat plate facing the airflow [30]. Many other studies have shown that, with perforations, the temperature at fin tip was lower as compared to without perforation. And perforations not only reduced the amount of material used but also reduced the pressure drop across the heat sink.

Heat sinks can be manufactured using conventional manufacturing processes like casting, forging or machining (milling). Heat sinks increase the rate of heat dissipation at the cost of added pressure drop. To reduce the pressure drop, various geometric features can be incorporated in the heat sink structure [31]. Mostly the manufacturing scale of such features is too small, or the geometry is too complicated for conventional manufacturing, therefore additive manufacturing is used. Tseng et al. [32] developed heat sinks with different geometric features and compared the performance with chevron-type corrugation. They manufactured four different bodies centered cubic (BCC) porous structures having homogeneous and non-uniform porosity. Such small geometric features were developed using Laser-powdered metal additive manufacturing with titanium alloy. Collins et al. [33] successfully fabricated channels in the heat sinks using direct metal laser sintering (DMLS). They also manufactured permeable membrane microchannels in heat sinks using additive manufacturing with AlSi10Mg alloy. Additive manufacturing of complex parts is cost-effective as compared to conventional manufacturing [34].

In the current research, two of the geometric features are combined in one heat sink. The proposed heat sinks contain longitudinal perforations (in the direction of airflow) and slots on sides of each fin, as shown in Figs. 1–3. These modifications are inspired by positive outcomes of creating perforations [21] or slots [15] on a plate-fin heat

sink. It is proposed that combining both geometric features in one heat sink resulted in an enhanced heat dissipation rate while reducing the pressure loss across the heat sink and minimizing the weight of heat sink, significantly.

Additive layer manufacturing (ALM) and specifically selective laser melting (SLM) can be used to manufacture the complicated features of heat sinks [35,36]. In the current research, heat sinks contain longitudinal perforation of 2 mm hydraulic diameter through the fins of 34 mm length and some features are as small as 1 mm. Therefore, it is preferable to manufacture heat sinks using SLM.

2. Numerical analysis

2.1. Problem specification

The thermal conductivity of the material AlSi10Mg used in the 3d printing of plate fin heat sinks is taken as 190 W/m K. Plane plate-fin heat sink (PPFHS) is modified into novel plate-fin heat sink-1 (NPFHS-1) and novel plate-fin heat sink-2 (NPFHS-2) to provide extra surface area for convection heat transfer. All fins have a height (H), length (L), and thickness (t) of 34 mm, 34 mm, and 4 mm, respectively. Every heat sink contains three fins attached to its 5 mm thick base and the spacing between 2 consecutive fins (s) of a heat sink is assumed to be 6 mm. The perforation size (t_p) is 2 mm, height (H_c) of a channel (including channel wall thickness) is 4 mm, and the thickness of the fin at the location of slots (t_s) is 2 mm.

NPFHS-1 has 6 longitudinal square perforations having a 2 mm $\times$ 2 mm cross-section each, while NPFHS-2 has 6 longitudinal circular perforations with 2 mm diameter each. Apart from perforations, these heat sinks also have slots on each fin, as described in Fig. 1, 2 and 3.

2.2. Conjugate heat transfer model

Thermal airflows through PPFHSs are numerically analyzed by means of the conjugate heat transfer model in which air enters at 23 °C, and its velocity (U_{air}) ranges from 2 m/s to 8 m/s, and Reynold's number (Re) ranges from 13,049 to 52195, calculated by Equation (1). Hydraulic diameter (D_h) of the duct is considered as characteristic length for Re number calculation, as shown in Equation (2).

$$\text{Re} = \frac{\rho_{\text{air}} U_{\text{air}} D_h}{\mu_{\text{air}}} \quad (1)$$

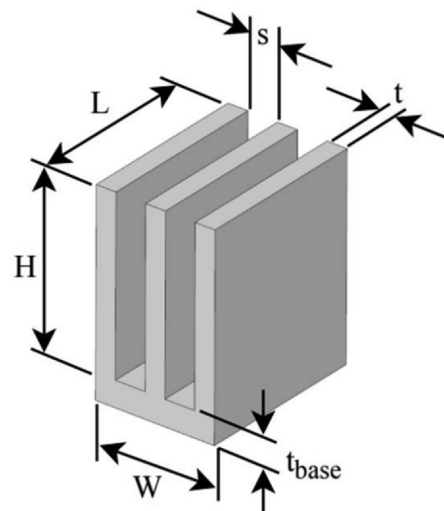


Fig. 1. Plane Plate Fin Heat Sink (PPFHS) CAD model.

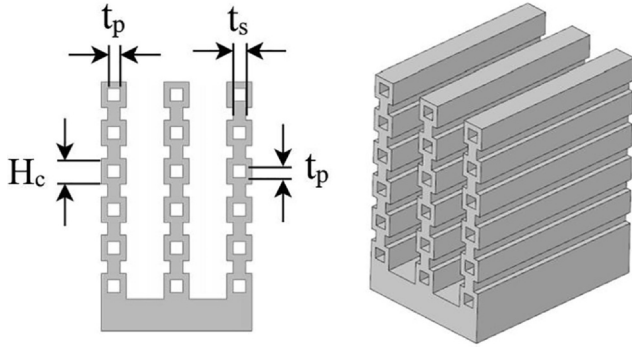


Fig. 2. Novel Plate Fin Heat Sink-1 (NPFHS-1) CAD model.

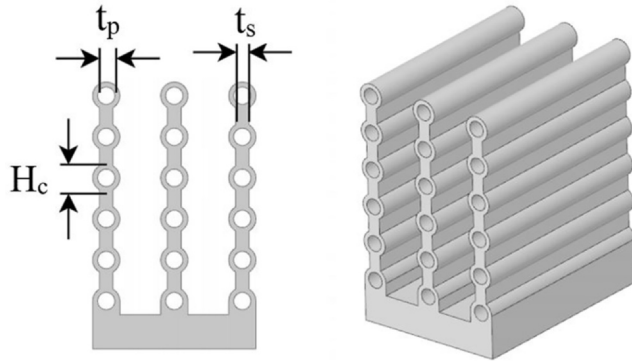


Fig. 3. Novel Plate Fin Heat Sink-2 (NPFHS-2) CAD model.

$$D_h = \frac{4A_c}{P} \quad (2)$$

where, A_c and P are the cross-sectional area and perimeter of the duct, while ρ_{air} and μ_{air} are density and dynamic viscosity of air. Steady, incompressible, and turbulent airflow is considered. Heat conducted through Aluminum alloy heat sink is balanced by the heat convected through the interface of the heat sink and air. Such a model is called the conjugate heat transfer model, as shown in Fig. 4.

The conduction of heat through aluminum alloy heat sink is solved using Fourier's law of conduction, Equation (3).

$$\nabla \cdot (k_s \nabla T_s) = 0 \quad (3)$$

where k_s is the conduction coefficient of AlSi10Mg, i.e., 190 W/m.K and T_s is the temperature field in the aluminum alloy heat sink.

Turbulent flow through the heat sink is modeled using Reynolds-Averaged Navier-Stokes (RANS) Equations (4 and 5). These equations are obtained by time-averaging the continuity, momentum, and energy equations and decomposing variables into mean and fluctuating components. RANS equations are used by other researchers as well, e.g., Zhou and Catton [37] and Leung and Probert [38].

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \underline{\underline{U}}) = 0 \quad (4)$$

$$\frac{\partial (\rho \underline{\underline{U}})}{\partial t} + \nabla \cdot (\rho \underline{\underline{U}} \underline{\underline{U}}) = \nabla \cdot (\underline{\underline{\sigma}} - \rho \underline{\underline{U}}' \underline{\underline{U}}') \quad (5)$$

where $\underline{\underline{\sigma}} = -p\underline{\underline{I}} + \mu(\nabla \underline{\underline{U}} + |\nabla \underline{\underline{U}}|^T)$ and $-\rho \underline{\underline{U}}' \underline{\underline{U}}' = \mu_t (\underline{\underline{U}} + |\underline{\underline{U}}|^T) - \frac{2}{3}(\rho k \underline{\underline{I}})$ are Newtonian and Reynolds stress tensors respectively. While p , ρ , and μ are pressure density and viscosity of air, respectively. $\underline{\underline{U}}$ and $\underline{\underline{U}}'$ are average and turbulent fluctuation velocity vectors. $\underline{\underline{I}}$ is a unit tensor.

The temperature field in the fluid (T_f) is calculated by solving the RANS equations with the energy Equation (6).

$$\rho \frac{\partial C_p T_f}{\partial t} + \rho \underline{\underline{U}} \cdot \nabla (C_p T_f) = \nabla \cdot \left(k + \frac{C_p \mu_t}{Pr_t} \right) \nabla T_f + \dot{Q} \quad (6)$$

where $\dot{Q}$, C_p , and Pr are heat source (watts), the specific heat capacity of air and Prandtl number of air, respectively. The subscript t indicates turbulent counterparts.

To model thermal airflow through the heat sink, k - ω SST model, Equations (8–14), is used. This model has been used by Zhou & Catton [37], Leung & Probert [38], and many other researchers to predict highly separated flows. It combines k - ω in near-wall regions and k - ϵ at the far-field of the free stream to precisely predict turbulent thermal airflow through the aluminum alloy heat sink. In this model, automatic wall function treatment is used, and heat transfer due to radiation is neglected. Change in the air properties due to temperature is not considered. The values of air density and viscosity are supposed to be constant and equal to the ones at the inlet, at 23 °C.

$$\frac{\partial (\rho k)}{\partial t} + \underline{\underline{U}} \cdot \nabla (\rho k) = \tilde{P}_k - \beta \rho k \omega + \nabla \cdot [(\mu + \sigma_k \mu_t) \nabla k] \quad (7)$$

$$\begin{aligned} \frac{\partial (\rho \omega)}{\partial t} + \underline{\underline{U}} \cdot \nabla (\rho \omega) = & \alpha \rho S^2 - \beta \rho \omega^2 + \nabla \cdot [(\mu + \sigma_\omega \mu_t) \nabla \omega] \\ & + 2(1 - F_1) \rho \sigma_{\omega_2} \frac{1}{\omega} \nabla k \cdot \nabla \omega \end{aligned} \quad (8)$$

where F_1 is blending function.

$$F_1 = \tanh \left(\min \left[\max \left(\frac{\sqrt{k}}{\beta^* \omega y}, \frac{500\nu}{y^2 \omega} \right), \frac{4\rho \sigma_{\omega_2} k}{CD_{k\omega} y^2} \right] \right)^4 \quad (9)$$

In which,

$$CD_{k\omega} = \max \left(2\rho \sigma_{\omega_2} \frac{1}{\omega} \nabla k \cdot \nabla \omega, 10^{-10} \right) \quad (10)$$

The turbulent eddy viscosity is defined by,

$$\nu_t = \frac{a_1 k}{\max(a_1 \omega, SF_2)} \quad (11)$$

where S and F_2 indicate invariant strain rate and a second blending function, respectively.

$$F_2 = \tanh \left(\left[\max \left(2 \frac{\sqrt{k}}{\beta^* \omega y}, \frac{500\nu}{y^2 \omega} \right) \right]^2 \right) \quad (12)$$

In stagnation regions, a production limiter (P_k) is used to limit the growth of turbulence.

$$P_k = \mu_t \frac{\partial u_i}{\partial x_j} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \rightarrow \tilde{P}_k = \min(P_k, 10\beta^* \rho k \omega) \quad (13)$$

The constants in this SST model are $\beta^* = 0.09$, $\alpha_1 = \frac{5}{9}$, $\beta_1 = \frac{3}{40}$, $\sigma_{k1} = 0.85$, $\sigma_{\omega1} = 0.5$, $\alpha_2 = 0.44$, $\beta_2 = 0.0828$, $\sigma_{k2} = 1$ and $\sigma_{\omega2} = 0.856$.

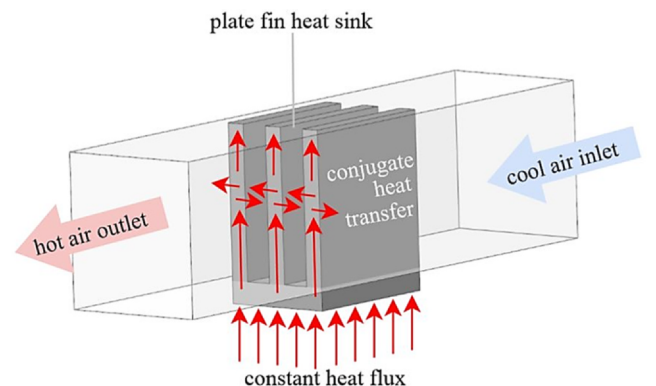


Fig. 4. Conjugate Heat Transfer Model of the plate-fin heat sink.

2.3. Computational domain and boundary conditions

Computational domain for numerical analysis is constructed in a way that entrance and exit walls are far enough from heat sink that reverse flow is avoided. The computational domain is shown in Fig. 5.

Thermal and flow boundary conditions for analysis are:

1. At the inlet, air temperature (T_{in}) is 23 °C while the velocity range is $2 \text{ m/s} \leq U_{air} \leq 8 \text{ m/s}$.
2. At the outlet, air pressure, $P = P_{gauge} = 0 \text{ Pa}$ and $\frac{dT}{dx} = 0$
3. At the top wall, $U_{air} = 0$, and $\frac{dT}{dz} = 0$
4. At right and left walls, $\frac{dU}{dy} = 0$ and $\frac{dT}{dy} = 0$
5. At the base of the heat sink, there is a no-slip condition ($U_{air} = 0 \text{ m/s}$) and a constant uniform heat flux of $Q = 8578.4 \text{ W/m}^2$.
6. At air-heat sink interface, there is no-slip condition ($U_{in} = 0 \text{ m/s}$) and heat flux is conserved. $k_f \cdot \frac{dT_f}{dn} = k_s \cdot \frac{dT_s}{dn}$

2.4. Solution methods and convergence criteria

Continuity is solved using the SIMPLE method. While, for coupled momentum and energy equations, the finite volume method-based code in ANSYS FLUENT is used. The solution is calculated using second-order upwinding. Flow characteristics are determined by solving continuity and momentum. Afterward, the energy equation is solved for the determination of the thermal characteristics of PFHSs. For the energy equation, the solution is considered converged when the sum of residuals is smaller than 10^{-6} . In the case of continuity and momentum equations, convergence criteria for relative residuals is 10^{-4} .

3. Experimental analysis

3.1. Airflow channel

The airflow channel is a close acrylic duct of 1 m length and 5 mm wall thickness. The channel inlet has a flow straightener of hexagonal cells, for uniform distribution of flow on the heat sink. After the flow straightener, the test section of uniform internal cross-section starts, which contains the heat sink. At the heat sink base, a heating pad of 7 W is attached. After the test section, the airflow channel diverges to accommodate a DC suction fan of diameter 300 mm, at the outlet. The fan is fitted into the channel using custom brackets, 3D printed by Ultimaker 3. The speed of this 12 V fan is controlled by varying the input current that changes the power of the fan accordingly ($P = VI$). Eventually, airspeed can be varied up to a maximum of 8 m/s in the test section. An HT-81 anemometer measures the airspeed in the test section, as shown in Fig. 7. The airflow channel is equipped with

thermocouple insertion points. For data acquisition, WinDAQ DI-1000TC data logger is used, which feeds the temperature readings to a computer. Fig. 6 shows the schematic diagram of experimental system.

3.2. Heat sink and test section development

The process used to manufacture the heat sinks (PPFHS, NPFHS-1 and NPFHS-2) is Selective Laser Melting (SLM), an Additive Manufacturing system using powder materials and lasers for material processing. The material used for the process was AlSi10Mg powder. The metal powder is deposited layer by layer and a fiber laser is used to melt the powder particles. All the heat sinks were produced at the same time with the same printing parameters. We can confidently expect their internal roughness to be the same. The developed models with attached thermocouples of PFHSs are shown in Fig. 8, Fig. 9 and Fig. 10.

The measurement of main dimensions was done using a digital Vernier calipers as shown in Figs. 11–13. The dimensions of the 3D printed heat sinks are approximately same as the CAD models.

The test section is detachable from the airflow channel, for operational ease. It has a uniform internal cross-section of 100mmx100mm and a wall thickness of 5 mm. Ultimaker 3 is employed to manufacture the test section, using Fused Deposition Modelling (FDM) process. Heat sinks are held in place (one by one) on the 7-watt Kapton film heater with the help of thermally conductive epoxy. This flexible polyimide (PI) film heater uniformly heats up when current flows through its flat thin metal (a nickel-based alloy) wire of high resistance.

Calibrated J-type thermocouples are used to sense the temperatures at different points of the heat sinks. One thermocouple is attached to each one of the three fin-tips, and one thermocouple is attached to the base of the heat sink. Two thermocouples are sensing the ambient temperatures at the inlet and outlet of the test section. Thermocouples are fixed in place by making small indents (~1mm) on the heat sinks, and conductive epoxy is used for adhesion, as shown in Figs. 8–10. Figs. 14 and 15 show the airflow channel assembled with the test section that is attached to the data logger and then a computer.

3.3. Experimental measurements and data analysis

The temperature readings of fin tips, heat sink, test section inlet, and outlet are obtained from thermocouples. Then the heat transfer rate through the heat sink is identified, knowing the thermal conductivity of AlSi10Mg heat sink. heat transfer rate. The total heat transfer rate at steady state is calculated using Equation (14).

$$\dot{Q}_e = \dot{Q}_c + \dot{Q}_r + \dot{Q}_l \quad (14)$$

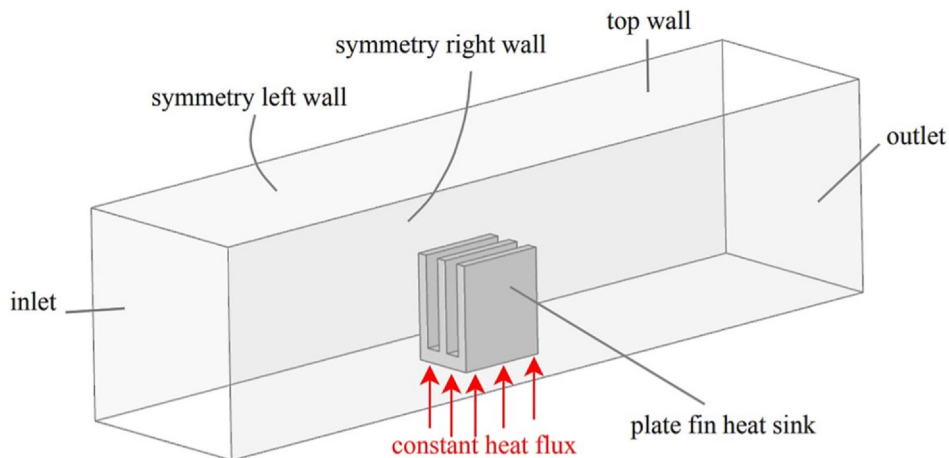


Fig. 5. Schematic diagram of the flow domain of CFD analysis with plate-fin heat sink.

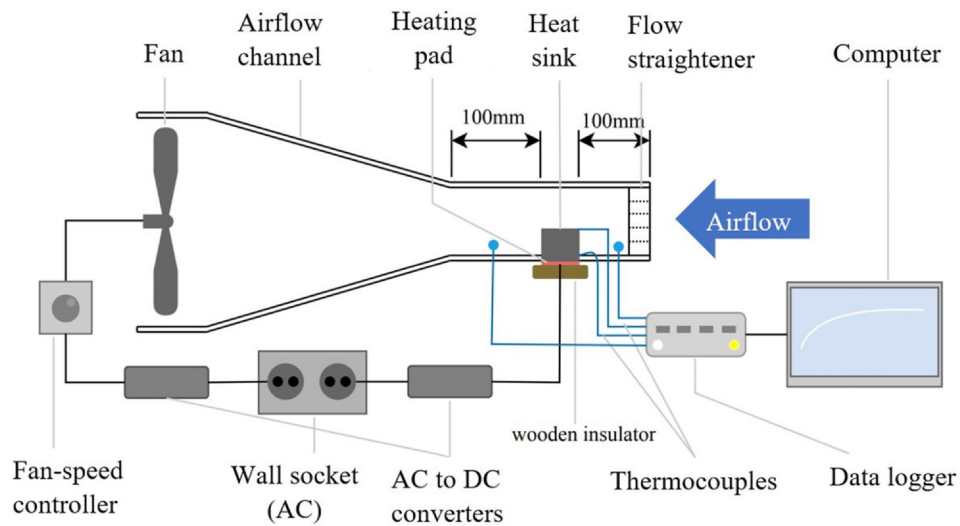


Fig. 6. Schematic diagram of the experimental setup.

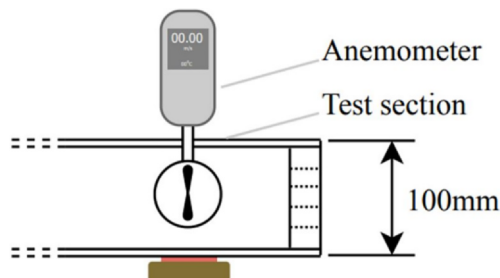


Fig. 7. Anemometer placement in the test section.

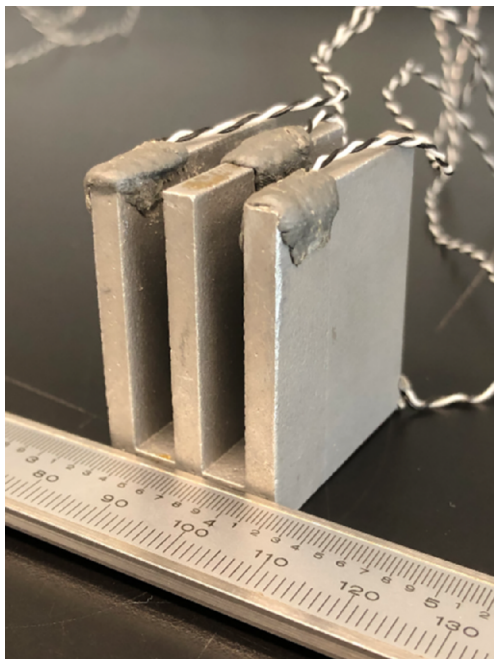


Fig. 8. Plain plate fin heat sink (PPFHS).

where, $\dot{Q}_e$ is the rate of heat transfer provided by the PI heater. $\dot{Q}_r$ is the heat transfer rate through the heat sink, because of radiation, given in Equation (15). For aluminum alloy heat sink, emissivity (ϵ) is low because it is highly polished.



Fig. 9. Novel plate fin heat sink-1 (NPFHS-1).

$$\dot{Q}_r = \sigma \in A_S(T_s^4 - T_a^4) \quad (15)$$

where σ is Stephen Boltzmann constant, A_s is the surface area of the heat sink, T_s is the average surface temperature of the heat sink surface, and T_a is the average bulk/ambient temperature around the heat sink. Experimental data depicts that the radiation heat transfer rate is low enough to be neglected. On average, for each heat sink, $\frac{Q_r}{Q_c}$ is less than 0.01.

Q_i is the heat transfer rate due to conduction through the insulation. If k_{ins} is the thermal conductivity of insulation, the conduction heat transfer can be calculated using Equation (16).

$$\dot{Q}_l = \frac{k_{\text{ins}} A_b}{\Delta x} \Delta T_{\text{ins}} \quad (16)$$

where, A_b is the area of heat sink base or the surface area of PI heater, and Δx is the thickness of the insulation. ΔT_{ins} is the temperature difference across the insulation. Experiments show that, on average, for each heat sink, $\frac{Q_1}{Q_0}$ is not more than 0.007, due to which conduction loss



Fig. 10. Novel plate fin heat sink-2 (NPFHS-2).

is neglected.

The convection heat transfer coefficient (h_p) of the heat sink can be calculated using Equation (17).

$$h_p = \frac{\dot{Q}_c}{A_p \left[T_s - \frac{T_{out} + T_{in}}{2} \right]} \quad (17)$$

where, $\dot{Q}_c$ is the heat transfer rate due to convection. A_p is the projected area of the heat sink, also known as the surface area of the heat sink base. If w and l are width and length of the base, then $A_p = WL$. While T_s is the average temperature of the upper surface of the heat sink. T_{out} and T_{in} are the outlet and inlet air temperatures of the test section.

Nusselt number (Nu_p) can be calculated using the Equation (18), considering characteristic length (L) to be the length of the heat sink in the direction of airflow. Perforations and slots are made to modify the plate fins into novel fins. All other dimensions are same for both novel plate fins. The NPFHS-1 is modified form of PPFHS after removing some material in the form of longitudinal perforations and slots. Same goes for NPFHS-2 where only the shape of perforations and slots is different. The purpose is to compare the plain fins to its modified forms. And characteristic length of the heat sinks is taken as the length of the heat sink in the direction of flow, i.e. same for all heat sinks. Amer Al-Damook used the same characteristic length for plain and perforated pin-fins i.e. the length of heatsink along the direction of the fluid flow [39].

$$Nu_p = \frac{h_p L}{k_{air}} \quad (18)$$

where, k_{air} is the thermal conductivity of air.

4. Results and discussion

4.1. Grid independence test (GIT)

For numerical solution, heat sinks and computational domain are discretized into small cells (mesh). NPFHS-1 has hexahedral cells, while NPFHS-2 has tetrahedral mesh elements, as shown in Fig. 16 and Fig. 17. At a certain point, further refinement of mesh does not contribute to a significant change in the results. Mesh independence is supposed to be achieved when the difference between two consecutive values of ΔP and T_{base} is less than 0.5% for all PFHSs. The number of cells required to achieve mesh independence for PFHSs is listed in Table 1.

4.2. Validation of numerical method

Yang [40] performed experiments on the Aluminum plate-fin heat sink of thermal conductivity, $k = 170 \text{ W/m.K}$ under free stream velocities (U_{air}) of 1–5 m/s. There are 15 fins on a square base of $45\text{mm} \times 45\text{mm} \times 2\text{mm}$, while each fin is 1 mm thick. These experiments are simulated using our numerical model, and pressure drop values at

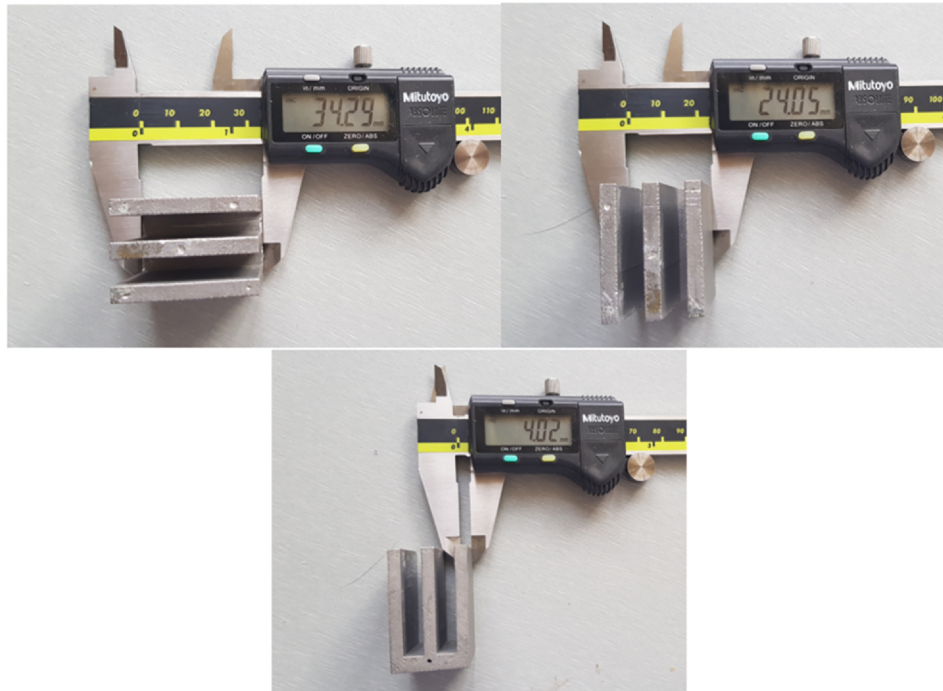


Fig. 11. Dimensional validation of the PPFHS features.

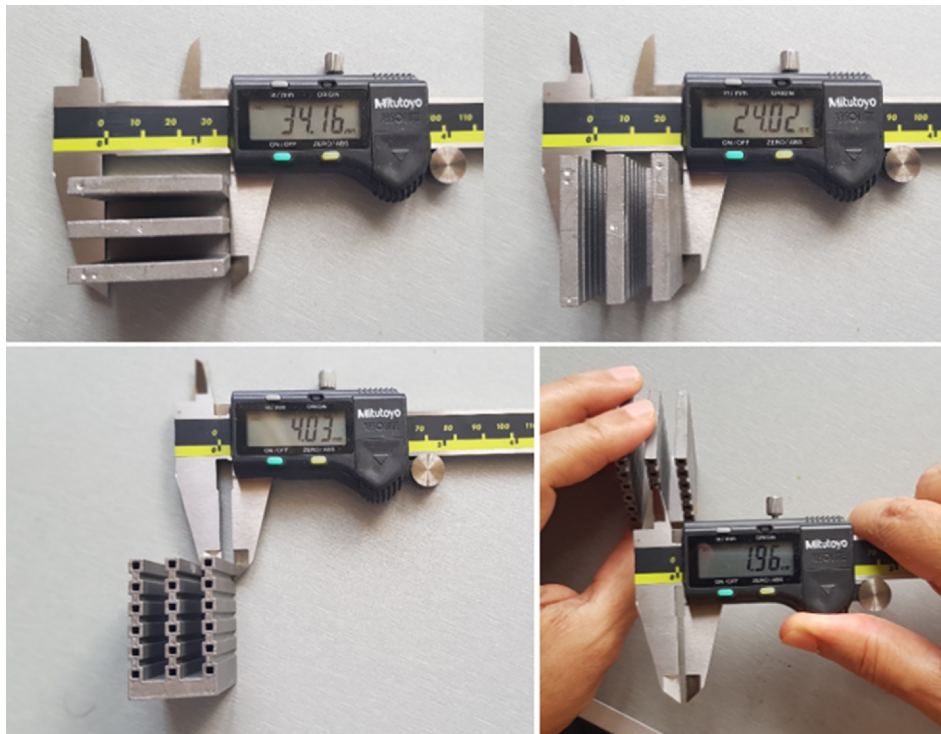


Fig. 12. Dimensional validation of NPFHS-1 features.

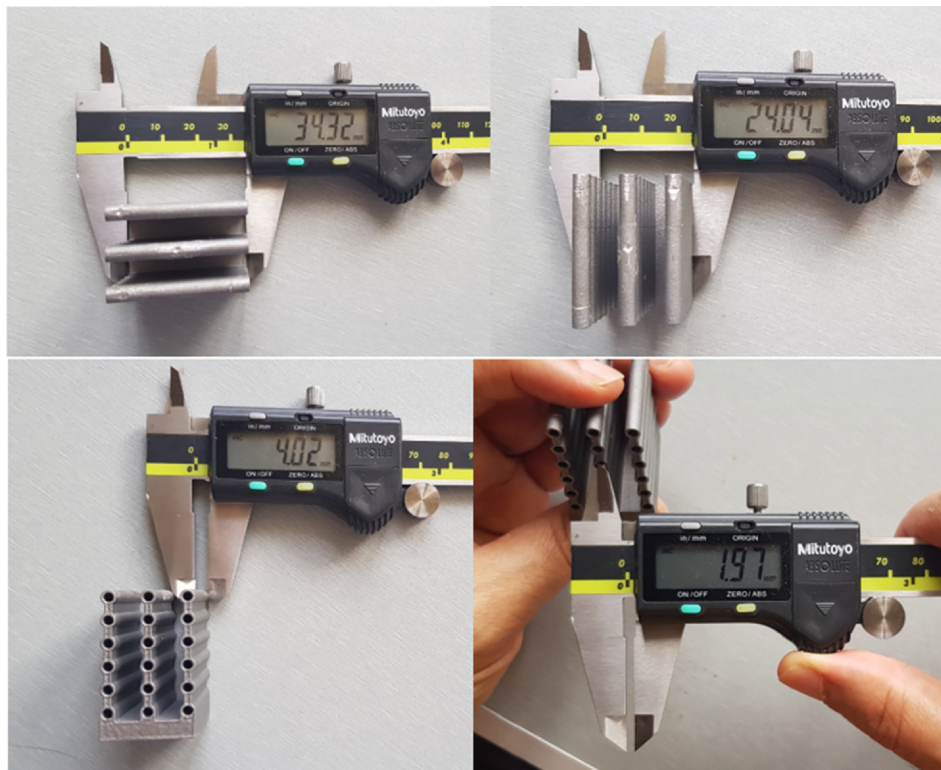


Fig. 13. Dimensional validation of NPFHS-2 features.

different air stream velocities are calculated and compared to the experimental pressure drop values. Fig. 18 shows strong agreement between computed and experimental data, with a maximum percentage error of 7.8%.

Convection coefficient (h_p) is calculated using temperature readings of the thermocouples at heat sink base and fin tips, at airflow speeds of

2 m/s to 8 m/s. These experimental values are compared to the h_p values calculated from the temperature values obtained from the numerical CFD model. Simulated and experimental h_p values of PPFHS, NPFHS-1, and NPFHS-2 are plotted against different velocities shown in Figs. 19–21, respectively. The maximum percentage difference between experimental and simulated values of h_p for PPFHS is 5.4%. While for

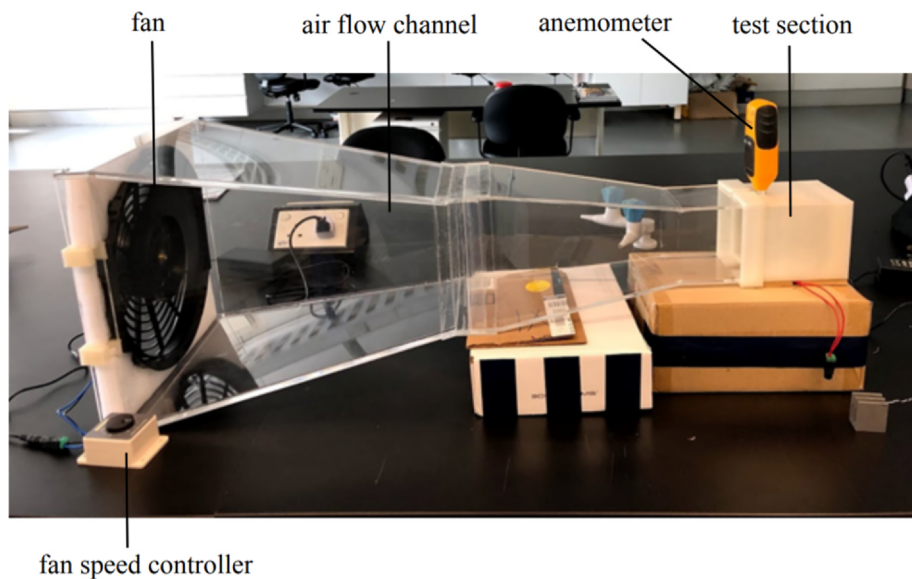


Fig. 14. Experimental setup: airflow channel with test section attached.

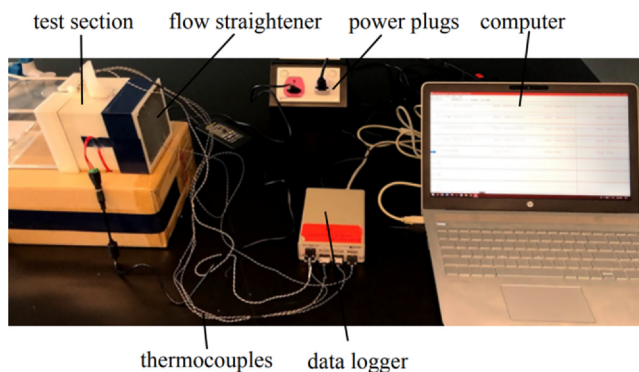


Fig. 15. Experimental setup: data acquisition setup.

NPFHS-1 and NPFHS-2, this difference is 5.7% and 5.3%, respectively.

4.3. Effect of novel fin-designs on heat transfer

NPFHS-1 and NPFHS-2 have the longitudinal square and circular perforations and slots in their fins. Because of these modifications in fin geometry, the heat transfer rate for novel fins is considerably higher than that of plain fins. Calculations show that, at considered air velocities, convection heat transfer coefficient (h_p) for NPFHS-1 and

NPFHS-2 is averagely 26.5% and 23.2% higher than that of PPFHS. Thus, for a given Re number, Nu number for NPFHS-1 is highest among all other heat sinks, and NPFHS-2 follows closely behind, as shown in Fig. 22. The improved performance of NPFHS-1 and NPFHS-2 is caused by increased surface area, exposed to heat transfer. Comparison between NPFHS-1 and NPFHS-2 shows that NPFHS-1 has 10.7% higher h_p and Nu number than NPFHS-2, because of relatively high surface area, as shown in Table 2.

In electronics, the heat sinks are placed on a surface where the temperature is intended to be reduced. Thus, a heat sink with a relatively lower base temperature (T_b) has higher effectiveness. On average, the values of T_b for NPFHS-1 and NPFHS-2 are 3.25% and 2.25% lower than those of PPFHS, as shown in Fig. 23. NPFHS-1 provides lowest base temperatures at all air speeds. This is because this heat sink, with square perforations and slots, has surface area 10.3% and 42.2% higher than that of NPFHS-2 and PPFHS. Temperature distribution profile in each heat sink shows that temperature on the side facing air-inlet, is relatively lower than the other side of heat sink. Figs. 24–26 show the temperature distribution in plate fin heat sinks at a Re number 13049. The temperature of the forward side of the heat sink is relatively lower than that of the trailing end of the heat sink. Possible reason for such temperature distribution is increase in the thickness of thermal boundary layer along the flow direction.

The Nu number of plate fins are correlated to the Re number as

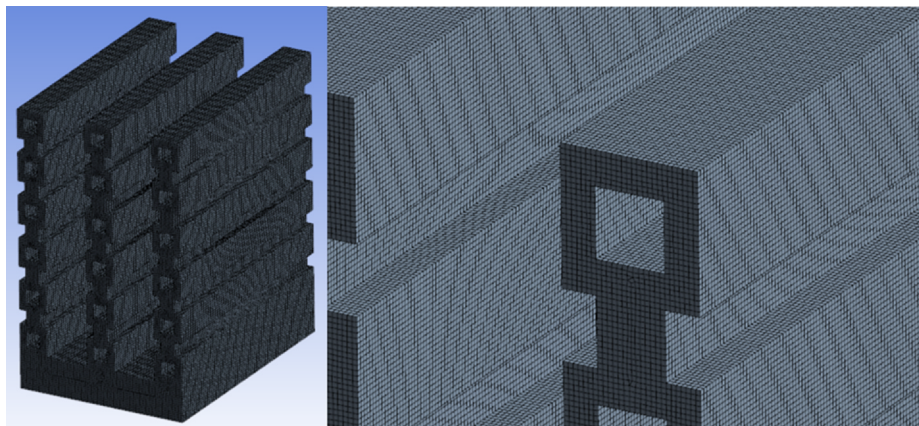


Fig. 16. Hexahedral cells of NPFHS-1.

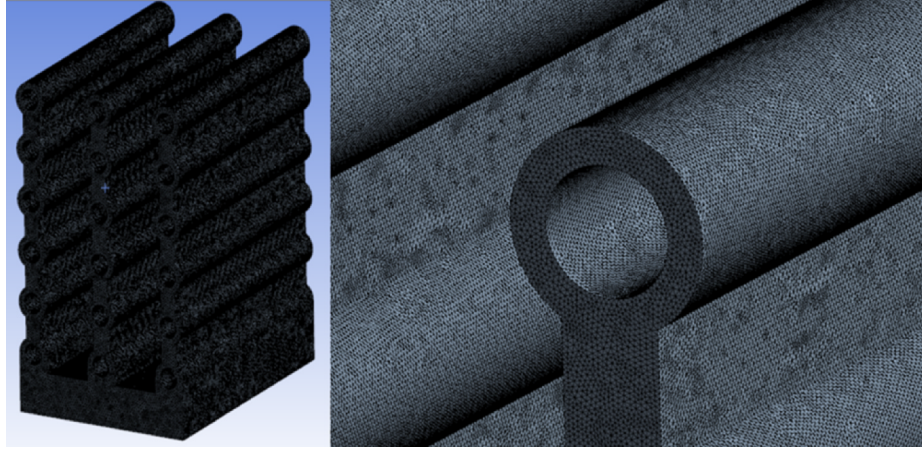


Fig. 17. Tetrahedral cells of NPFHS-2.

Table 1
Grid Independence Test.

PFHS type	cells	T_{base} (°C)	ΔP (Pa)
PPFHS	159,341	43.9	2.47
	231,753	45.5	2.52
	273,903	45.7	2.53
NPFHS-1	970,485	41.9	2.38
	1,253,348	44.1	2.41
	1,405,725	44.3	2.40
NPFHS-2	885,121	42.6	1.89
	1,134,052	44.4	2.04
	1,284,726	44.6	2.05

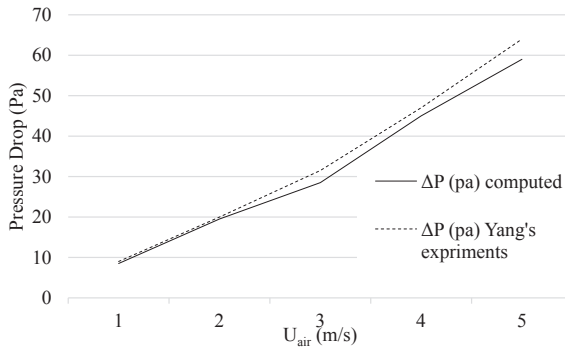


Fig. 18. Comparison of computed and experimental values of pressure drop.

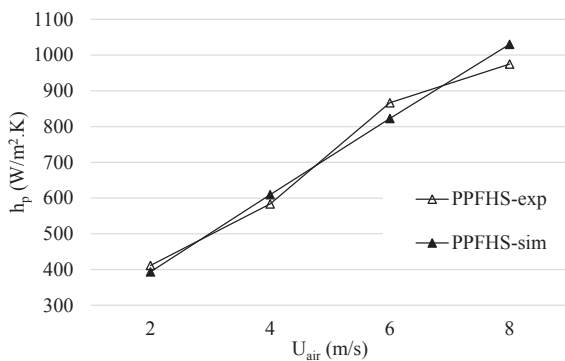


Fig. 19. Comparison between experimental and simulated results of the convection coefficient for plain plate-fin heat sink (PPFHS).

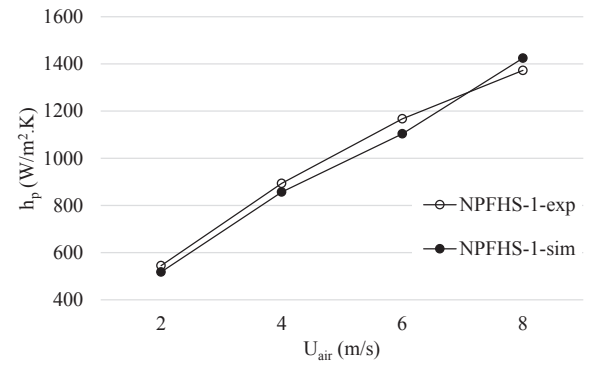


Fig. 20. Comparison between experimental and simulated results of the convection coefficient for novel plate-fin heat sink-1 (NPFHS-1).

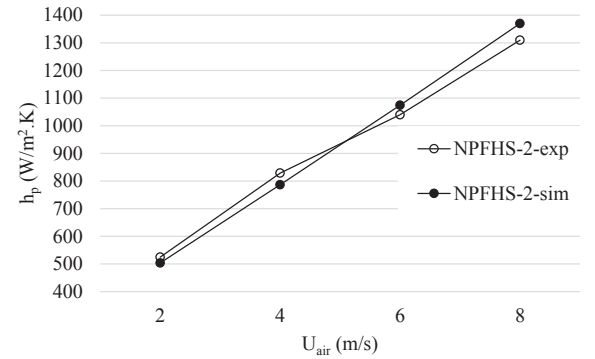


Fig. 21. Comparison between experimental and simulated results of the convection coefficient for novel plate-fin heat sink-2 (NPFHS-2).

shown in equations below:

$$\text{PPFHS: } Nu = -2.7144Re^2 + 290.15Re + 225.01$$

$$\text{NPFHS-1: } Nu = 673Re^{0.7178}$$

$$\text{NPFHS-2: } Nu = 4.084Re^2 + 355.58Re + 296.68$$

4.4. Effect of novel fin-design on pressure drop

A heat sink placed in the airflow path causes a reduction in air pressure. This pressure-drop (ΔP) significantly depends on the geometry of the heat sink. NPFHS-1 and NPFHS-2 contain perforations and slots in the direction of flow, reducing the resistance in the flow path. Pressure drop across heat sink is calculated using Equation (19).

$$\Delta P = P_{out} - P_{in} \quad (19)$$

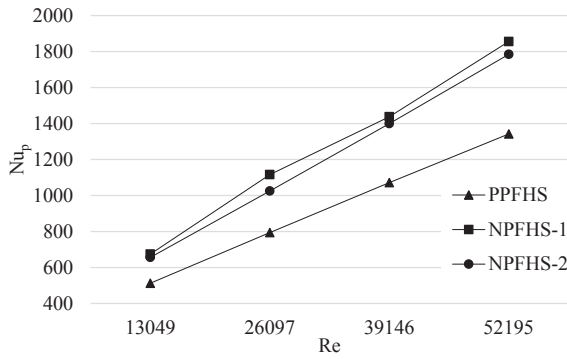


Fig. 22. Nu numbers of all plate-fin heat sinks at different Re numbers.

Table 2

Surface areas of all plate-fin heat sinks.

A_s (mm ²)	PPFHS	NPFHS-1	NPFHS-2
	9148	15,820	14190.3

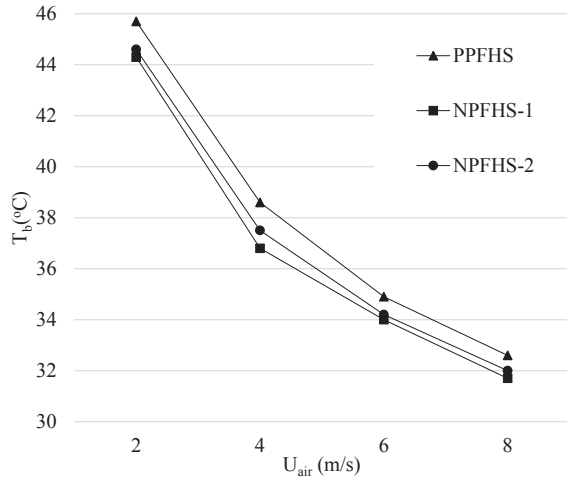


Fig. 23. Temperatures of heat sink bases at different airspeeds.

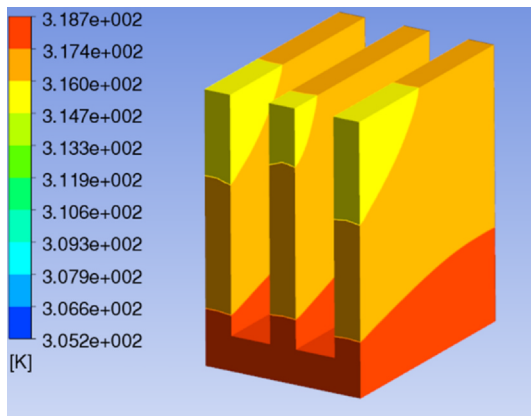


Fig. 24. Temperature distribution in PPFHS at Re = 13049.

The CFD model shows that NPFHS-1 and NPFHS-2 cause 29.1% and 41.1% lesser pressure-drop than that of PPFHS. Thus, NPFHS-2 offers the least hindrance to airflow, while PPFHS offers the most hindrance, as shown in Fig. 28. Air pressure profiles around the heat sinks at a Re number of 13049, are shown in Figs. 30–32. Pressure is dropped as the air passes over the heat sink. NPFHS-2 causes the least pressure to drop

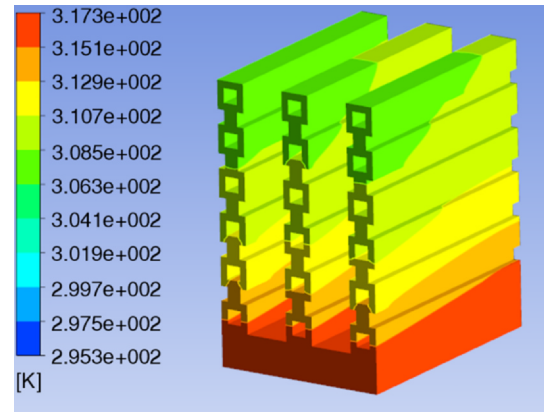


Fig. 25. Temperature distribution in NPFHS-1 at Re = 13049.

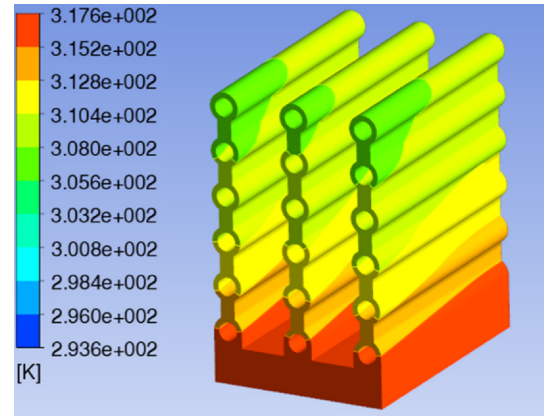


Fig. 26. Temperature distribution in NPFHS-2 at Re = 13049.

while PPFHS causes the most pressure drop.

NPFHS-2 performs better than the other two in terms of lower pressure drop across it. Because the smallest frontal area causes the least friction in the path of airflow, as shown in Fig. 27 and Table 3.

When pressure drop across a heat sink is higher, more power is needed to push the air through the heat sink. To overcome the pressure-drop, required fan power (p_{fan}) is highest for PPFHS and lowest for NPFHS-2, as shown in Fig. 29. Pumping power increases when the hindrance in the path of flow is enlarged. But in the case of NPFHS-1 and NPFHS-2, longitudinal perforations and slots cause the frontal area of the fins to be reduced, as shown in Fig. 27. Thus, more total surface area is achieved while having less interruption to the flow. This causes lesser pressure to drop and eventually lesser fan power is required.

If A_f is the area available for air to flow through the heat sink, then p_{fan} can be calculated using Equation (20) [39].

$$p_{fan} = U_{air} \cdot A_f \cdot \Delta P \quad (20)$$

The relation between friction factor and pressure drop is $f = \frac{\Delta P}{0.5 \rho U_{air}^2 \frac{L}{D_h}}$ [41,42]. Where ρ and U_{air} are the density and velocity of air. L is the distance between the points between which pressure drop is measured, while D_h is the hydraulic diameter of the flow channel. For the PPFHSs the friction factor is correlated to the Re as shown in Equations below:

$$\text{PPFHS: } f = 0.0055Re^2 - 0.05533Re + 0.3421$$

$$\text{NPFHS-1: } f = 0.2172Re^{-0.29}$$

$$\text{NPFHS-3: } f = 0.1852Re^{-0.325}$$

4.5. Mass of heat sinks

Mass of a heat sink is also significant, especially when it comes to

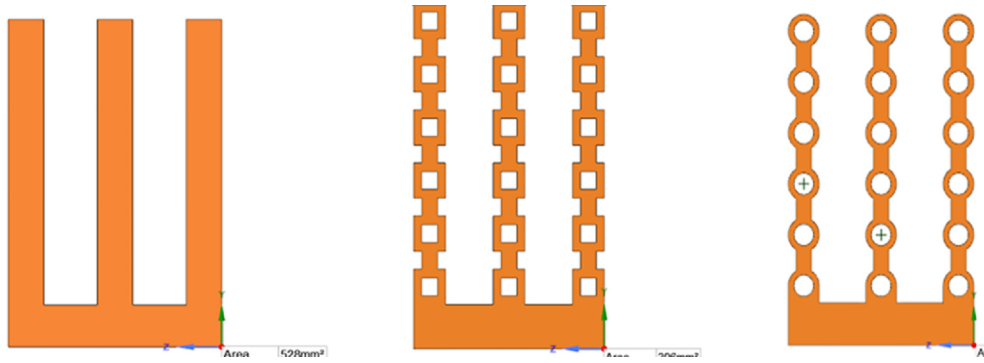


Fig. 27. Frontal areas of the PFHSs.

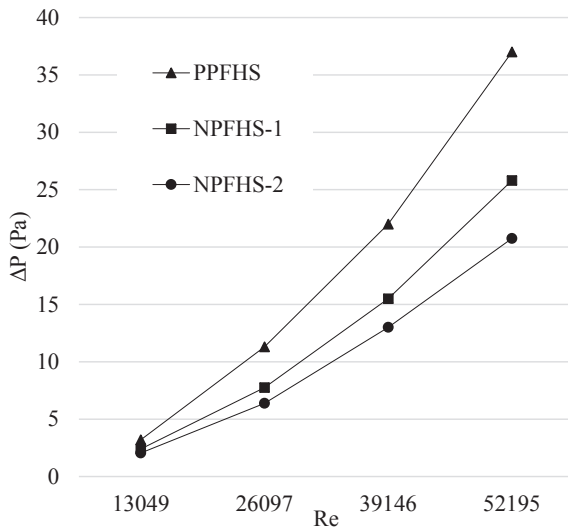


Fig. 28. Pressure drop caused by plate-fin heat sinks at different Re numbers.

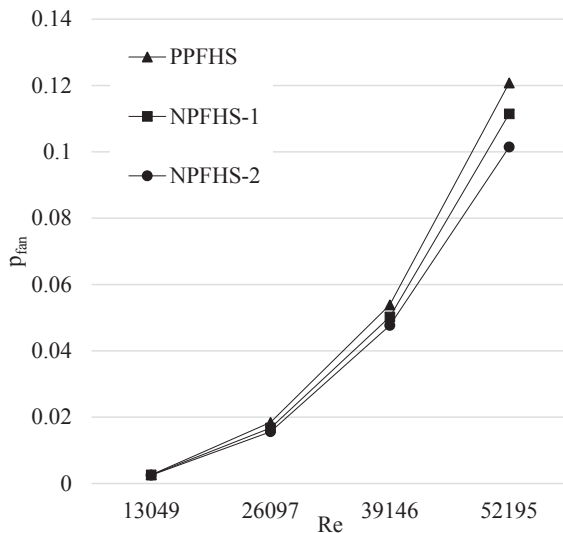


Fig. 29. Fan power required for the plate-fin heat sinks at different Re numbers.

the application in electronics. Lightweight heatsinks provide a cost-effective solution to tackle overheating. As NPFHS-1 and NPFHS-2 have perforations and slots, therefore their masses are less than the mass of PPFHS. Mass of the heat sink increases when some material is added to modify the geometry. But in current case, modification in plain fins is done by removing the mass in airflow direction by making perforations

and slots. This causes the NPFHS-1 and NPFHS-2 to be lighter in weight than the PPFHS.

If ρ_{Al} is the density of aluminum alloy and V_{hs} is the total volume of the heat sink, the mass of a heat sink (m) can be calculated using Equation (21).

$$m = \rho_{Al} (V_{hs}) \quad (21)$$

where the total volume of the heat sink is the sum of the volume of the base (V_{base}) and fins (V_{fins}), as shown in Equation (22).

$$V_{hs} = V_{base} + V_{fins} \quad (22)$$

NPFHS-1 and NPFHS-2 have 25% and 38% less mass than PPFHS. Total masses of all heat sinks are shown in Table 4.

5. Conclusion

The process of heat transfer occurs when there is a thermal gradient between the two media. The rate of heat transfer plays a critical role in engineering applications, and heat sinks are used to enhance it. Plate fins are commonly used heat sink in electronics and automobile engines. Various geometries of plate fins, having more surface area, are proved to increase the heat transfer rate. In the current study, the effect of perforations and slots through the plate fins is analyzed. Four novel fins are proposed with rectangular and circular shaped perforations and slots, named NPFHS-1 and NPFHS-2, respectively. Thermal airflows through PFHSs are numerically analyzed using the conjugate heat transfer model. The heat sinks are also analyzed experimentally and a decent agreement between numerical and experimental results is found. It is concluded that:

- The combination of perforations and slots enhances the convection coefficient. For a Re number of 13,049 to 52195, The NPFHS-1 and NPFHS-2 have averagely 26.5% and 23.2% higher Nu number (Nu_p) than non-perforated and non-slotted plate-fin heat sink (PPFHS). NPFHS-1 has the highest heat transfer coefficient because of the most surface area exposed to convection.
- Longitudinal perforations and slots through the plate fins (NPFHS-1 and NPFHS-2) drop lesser pressure than the non-perforated PPFHS. The NPFHS-1 and NPFHS-2 averagely cause 29.1% and 41.1% lower pressure drop than PPFHS, requiring lower fan power to overcome the pressure drop. NPFHS-2 cause the least pressure-drop because of the smallest frontal area of the heat sink.
- Because of perforations and slots, NPFHS-1 and NPFHS-2 have 25% and 38% less mass than that of PPFHS. Thus, novel fins are relatively lightweight than plain fins.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

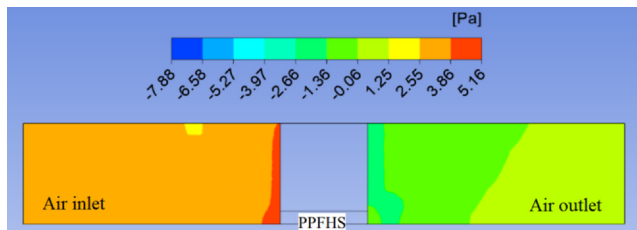
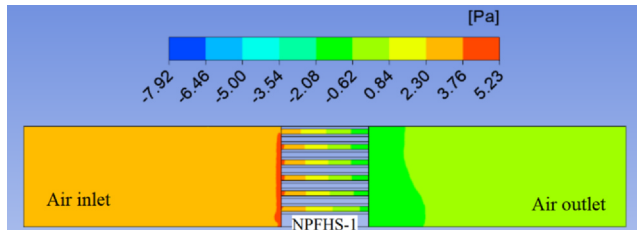
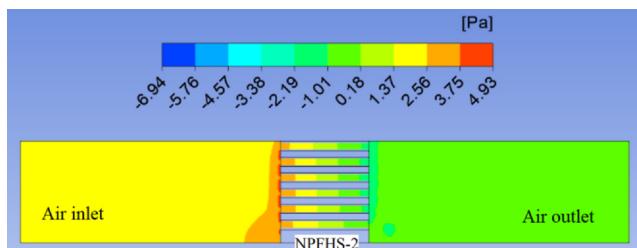
Fig. 30. Pressure distribution across PPFHS at $Re = 13049$.Fig. 31. Pressure distribution across NPFHS-1 at $Re = 13049$.Fig. 32. Pressure distribution across NPFHS-2 at $Re = 13049$.

Table 3
Frontal areas of the PFHSs.

Frontal Area (mm ²)	PPFHS	NPFHS-1	NPFHS-2
	528	396	325

Table 4
Masses of plate-fin heat sinks.

m (kg)	PPFHS	NPFHS-1	NPFHS-2
	4.87E-02	3.65E-02	3.00E-02

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2020.115949>.

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